Review of the Article:

Combining single-cell ATAC and RNA sequencing for supervised cell annotation.

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➢Why Annotation Matters?

Single-cell data reveals cellular **heterogeneity & function.**

A crucial step in single-cell analysis is annotation. Accurate **cell-type annotation** is essential for downstream analysis.

What is an annotation?

- Annotation is the process of **identifying and assigning cell types to single cells based on their molecular profiles (e.g., RNA or ATAC features).**
- FACS is the gold standard, but **not scalable** for thousands of cells
- Machine learning (unsupervised & supervised) enables rapid annotation

Current Practice

- Annotation typically relies on **RNA-seq** alone
- ATAC has been added in some **unsupervised** studies
- **Supervised use of ATAC** has not been deeply evaluated

Unsupervised Annotation

Unsupervised annotation involves clustering cells based on their sequencing profile, typically using RNA profiles.

- Similar expression between subtypes → poor separation
- Manual curation is labor-intensive
- Results not standardized → low reproducibility

Supervised Annotation

This method utilizes classification models for cell annotation. These models are trained on a reference dataset with a similar cell-type composition as the unannotated dataset.

- Fully data-driven
- Can use previously unknown discriminative features.

- RNA offers strong discrimination, but **ATAC is often unused**
- ATAC may capture epigenetic states missed by RNA, esp. **T-cell subtypes & heterogeneous tissues (e.g., cancers)**
- Motivate ATAC's potential to capture regulatory variation underlying cell states.



What is ATAC?

ATAC = Assay for Transposase-Accessible Chromatin

Uses **Tn5 transposase** to insert sequencing adapters into **open chromatin**

Sequencing reads identify **accessible regulatory regions** (promoters, enhancers)

What ATAC Measures

Chromatin accessibility → identifies where DNA is open

Reflects **regulatory potential**, transcription factor binding sites, and epigenetic state

Captures features **upstream of gene expression** (RNA)

Why ATAC Matters for Cell Annotation

Reveals **epigenetic signatures** that distinguish closely related subtypes

Provides complementary information not visible in RNA; Especially informative in **immune cells (e.g., T-cell states)** due to epigenetic "poised/bivalent" regions

Improvements in supervised annotation

- **Dimensionality reduction is crucial;** neural-network embeddings (e.g., scVI, GANs) capture complex non-linear relationships better than linear methods like PCA.
- **scVI improves latent representations** using a zero-inflated model and corrects batch effects, but neural networks may overfit noise in sparse single-cell data.
- **ATAC is more sparse than RNA,** making linear methods harder to apply, and non-linear classifiers (e.g., SVM) often perform slightly better for unimodal RNA or ATAC data.

- It is not yet clear whether **non-linear embeddings and classifiers** are required to fully exploit ATAC features.

- ATAC usefulness may be **tissue-dependent,** because epigenetic distinctions vary across tissues and across specific cell types within a tissue.
- Identifying where **ATAC provides meaningful signal helps determine when RNA+ATAC annotation** is necessary and when **RNA-only is sufficient**, reducing cost and complexity.

Methods

Study Objectives:

- ✓Compare **PCA vs scVI** and **linear vs non-linear classifiers** for RNA+ATAC
- ✓Determine whether ATAC improves supervised annotation
- ✓Assess differences between **immune vs neuronal tissues**
- ✓Evaluate ATAC utility across **cell types & label granularities**

Methods – Overview of the Multi-omic Cell Annotation Pipeline

1. Data Preparation

Peripheral blood or neuronal cells were profiled using **10x multi-omic sequencing**, generating matched **RNA** and **ATAC** modalities for each cell.

Standard **quality control (QC)** was applied to filter low-quality cells and features.

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2. Ground Truth Labelling

Features were **z-score normalized** using

to ensure comparable scales across modalities.

Weighted Nearest Neighbor (WNN) clustering was performed to integrate RNA and ATAC and identify cell clusters.

Clusters were annotated using **marker gene–based methods** to generate ground-truth labels.

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3. Supervised Annotation Framework

The dataset was split using **bootstrap sampling** into training and out-of-bag (OOB) test sets.

Pre-processed features (RNA, or RNA+ATAC) were used for **dimensionality reduction** via

PCA

scVI (deep generative embedding for multi-omics)

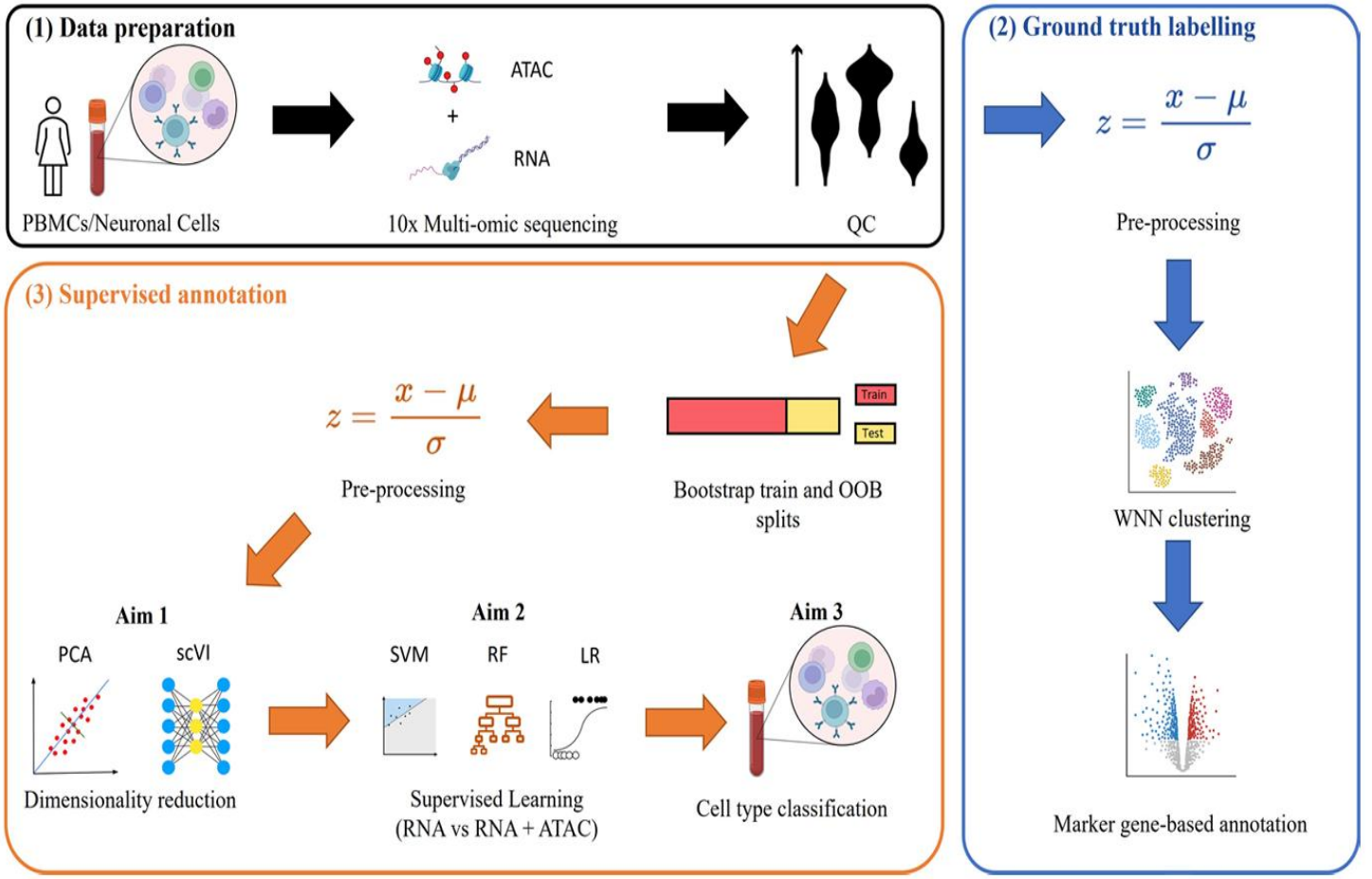
Classifiers were trained to predict cell types:

Support Vector Machine (SVM)

Random Forest (RF)

Logistic Regression (LR)

Models were evaluated on held-out cells to assess the accuracy of **cell type classification**.



Datasets

PBMCs: n = 11,909 cells (healthy female donor, age 25)
Neuronal: n = 10,530 cells (7 AD, 8 controls)
Both 10x multiome (RNA + ATAC)

Ground Truth Labeling

Here was no third-party dataset for providing a label for both RNA and ATAC (only RNA) → bias → self-generate labels → some subjective bias → minimized by using best practices for manual annotation → Multi-modal clustering.

- Clustering: PCA → WNN → Leiden
- Manual marker-based annotation
- Two label granularities: WNN-L1 (major), WNN-L2 (subtypes)
- 10 bootstrap splits for evaluation

Model Evaluation – Unsupervised Metrics

UMAP was applied to **PCA and scVI embeddings** to **visualize cell populations** in two dimensions using ground truth labels.

Silhouette scores were calculated using **Euclidean distances within the PCA and scVI embedded spaces** to **evaluate cluster separation quality.**

Silhouette scores were reported as **median and IQR** across 10 bootstrap training and out-of-bag test sets due to small sample size and non-normal score distribution.

Model evaluation

Supervised Metric: Proportion of Ambiguous Predictions (PAP)

PAP is a metric used to measure the **confidence** of supervised cell annotation models.
Supervised methods often leave low-confidence predictions **unannotated** if the cell type is not well represented in the training dataset. PAP quantifies how often the model makes **uncertain predictions**.

PAP is calculated using prediction probabilities from OOB test sets. It measures the proportion of predictions that fall between probability **0.1 and 0.9**, where predictions are ambiguous:

Lower PAP = higher model confidence and fewer unannotated cells.

- Predictions below 0.1 are very confident negative
- Predictions above 0.9 are very confident positive
- Predictions between 0.1 and 0.9 are ambiguous

Statistical Analysis

Non-parametric statistical tests were used because performance metrics did not follow **a normal distribution**.

Model comparisons evaluated **RNA + ATAC vs RNA-only**, since RNA-based models are known to outperform ATAC-only models.

The **Wilcoxon signed-rank test** was applied due to **paired bootstrap samples and violation of normality assumptions**.

Multiple testing correction was performed. Feature contributions were examined using **lasso regression**, and correlations between scVI components were assessed using **Pearson correlation coefficients**.

Result

ATAC Improves Cell Type Separation

Improved Embedding Discriminability

To evaluate the effect of ATAC features, **UMAP** was applied to **PCA** and **scVI** embeddings from bootstrap training samples.

Visual comparisons show **stronger separation** between cell populations when **ATAC + RNA** features are used.

PCA Embeddings

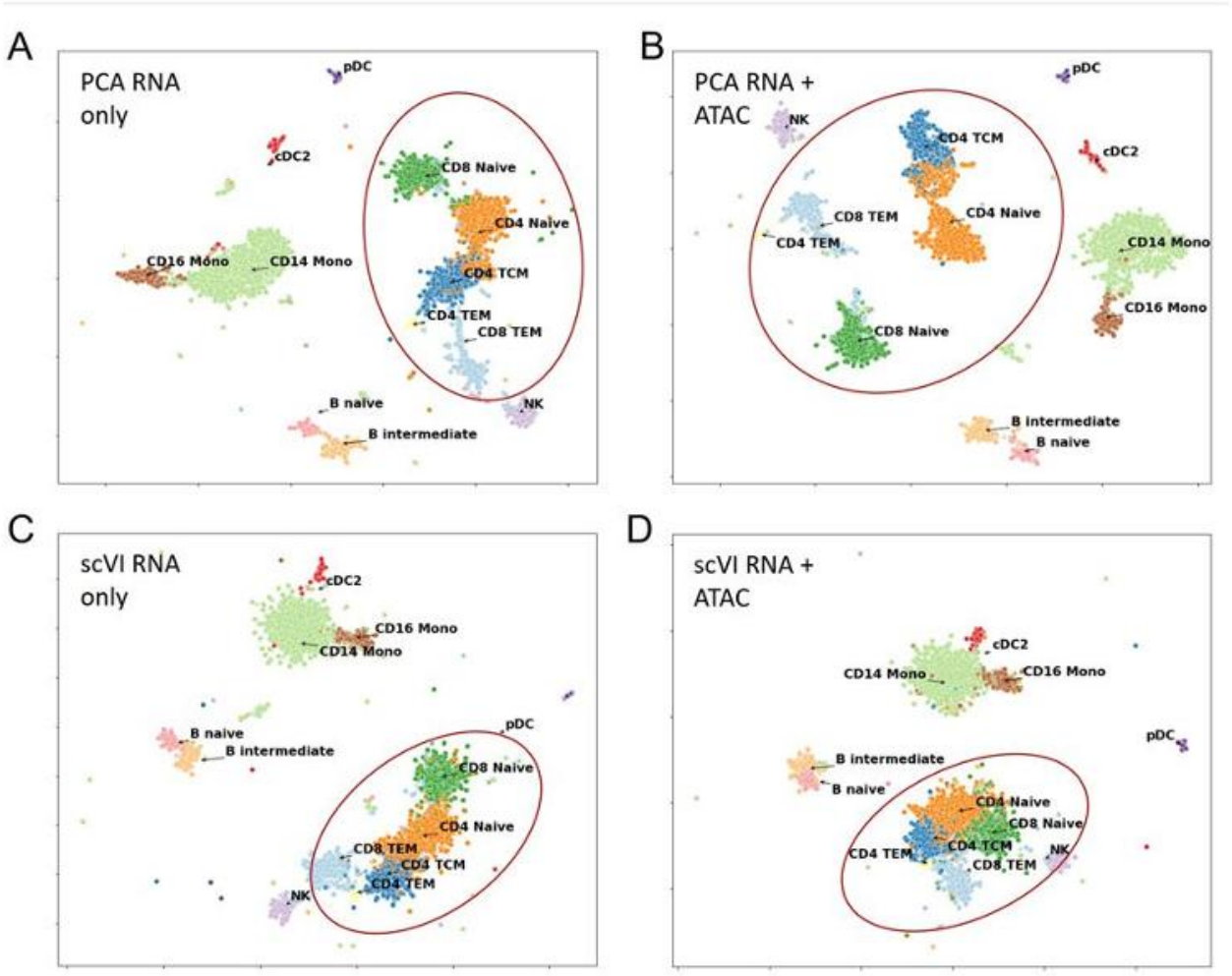
With ATAC: CD4 TEM, CD8 TEM, and CD8 naïve cells show clearer separation from CD4 naïve and CD4 TCM groups.

RNA only: These populations overlap more and are less distinguishable.

scVI Embeddings

Minimal visual difference between **RNA-only** and **combined RNA+ATAC** embeddings.

ATAC had limited effect on scVI's latent space structure.

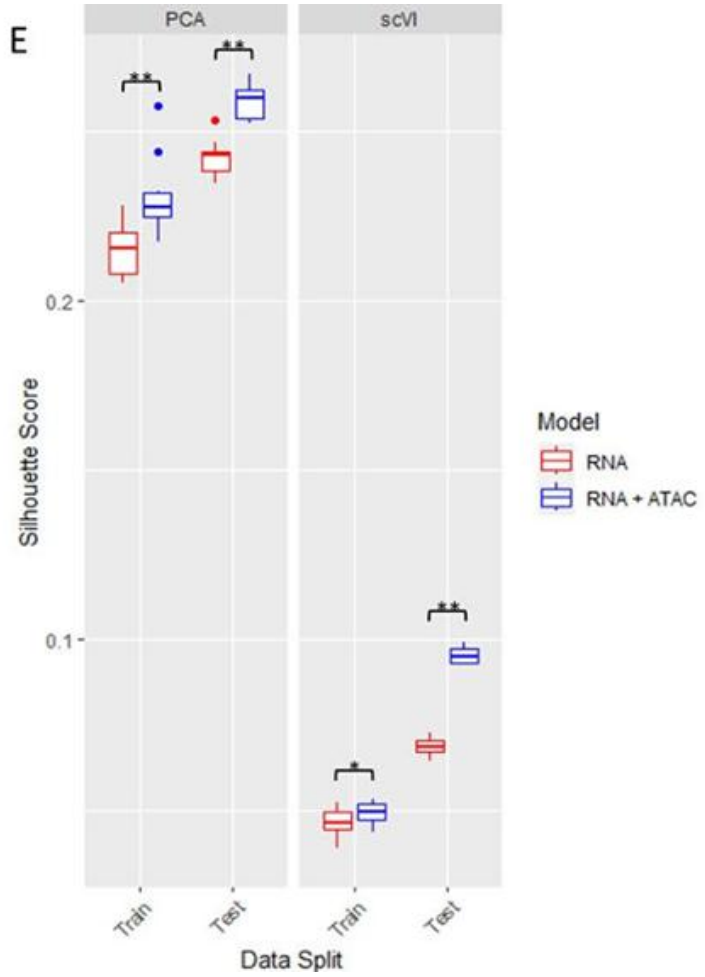


Quantitative Evaluation (Silhouette Scores)

Adding ATAC features **significantly increased silhouette scores** for both training and test sets in PCA and scVI.

Largest improvement was in the **scVI test set**:

RNA only: **median = 0.068** (IQR 0.067–0.071)
Combined: **median = 0.095** (IQR 0.093–0.098)
p < 0.01



ATAC Improves Classification in scVI Embeddings

Impact of Embedding Choice on Classification
•Macro F1 scores were compared across models using **RNA only** vs **RNA + ATAC**.

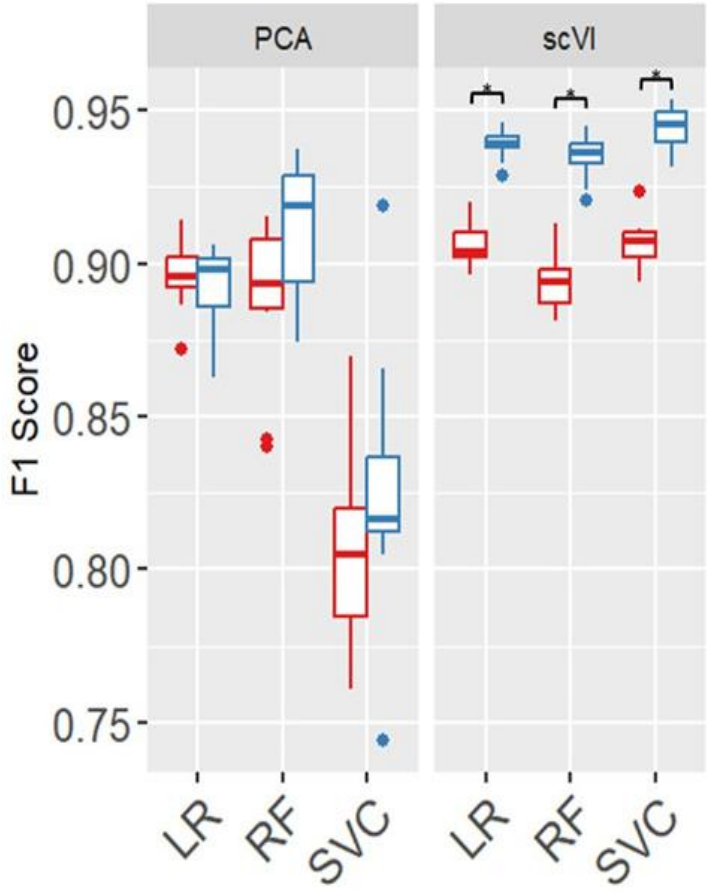
PCA Embeddings
•Showed **weaker classification performance** overall compared to scVI.
•Best PCA result: **RF with RNA + ATAC**

•Adding ATAC offered **minimal improvement** in PCA-based models.
•No significant differences in macro F1 between RNA-only and RNA+ATAC.

scVI Embeddings

•scVI models showed **consistently higher performance** than PCA.
•Best scVI performance: **RNA + ATAC**

•scVI showed **clear improvements** when ATAC was added for **every** classifier.
•Best-performing scVI classifier: **SVM + RNA + ATAC**, which significantly outperformed RNA-only



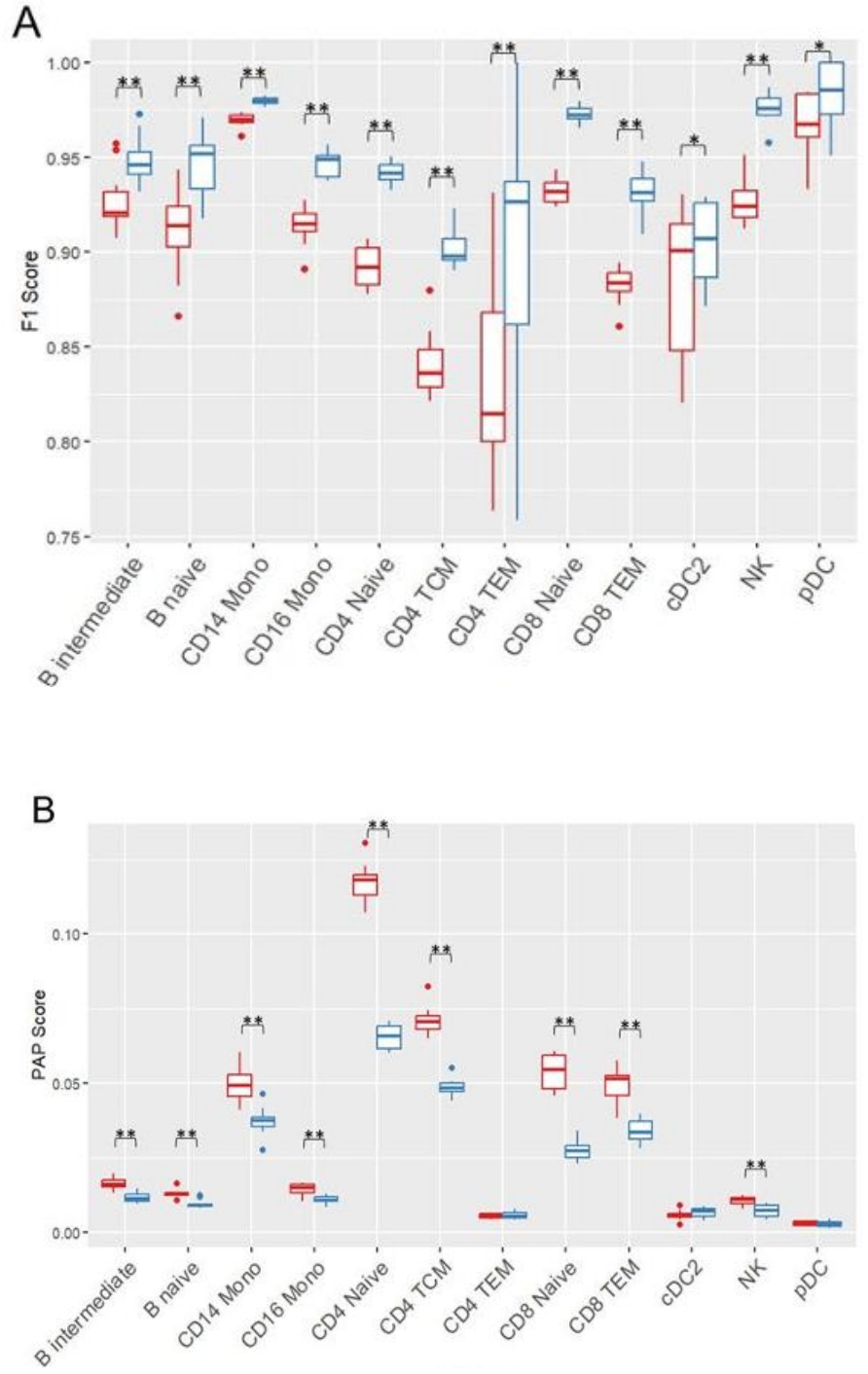
Cell-Type–Specific Effects of ATAC

Comparisons made between **RNA-only** vs **RNA + ATAC**.
F1 Score Improvements
ATAC improved F1 scores for every cell type.
T cell subtypes showed the strongest gains.
Largest improvement: **CD4 TEM**

PAP Score Reductions
T cells also showed the **largest reductions in PAP**, indicating reduced misclassification.
Example: **CD4 naïve**
Exception: **CD4 TEM** showed no significant PAP reduction.

Minimal Improvements in Dendritic Cells
Dendritic cell subtypes (pDC, cDC2) showed **smaller improvements**.

cDC2 F1 improvement was **very small but statistically significant.**



ATAC Helps Only for Fine-Grained Labels

Evaluated ATAC utility using **coarse (WNN-L1)** vs **granular (WNN-L2)** labels.

WNN-L1 (broad types):

Small precision increase with ATAC

No significant change in recall or F1

WNN-L2 (subtypes):

ATAC significantly improves precision, recall, and F1

- ATAC provides benefits **only for detailed subtype annotations**, not broad cell categories.

ATAC embeddings are associated with B and T cell subtypes

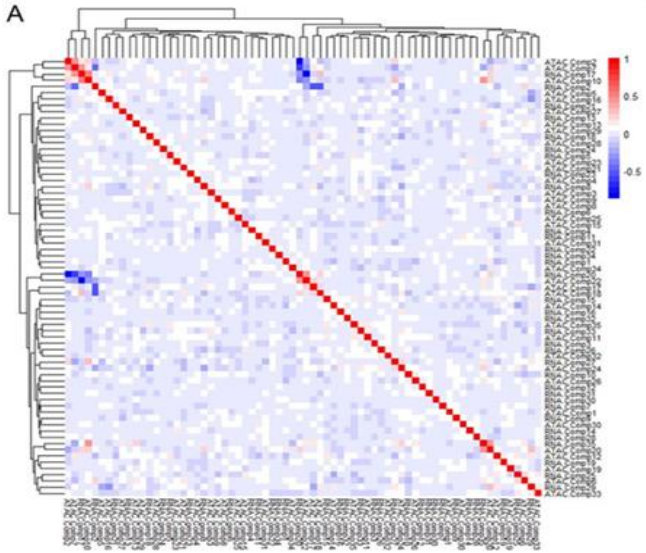
To understand how RNA and ATAC contribute to predictions, we analyzed:

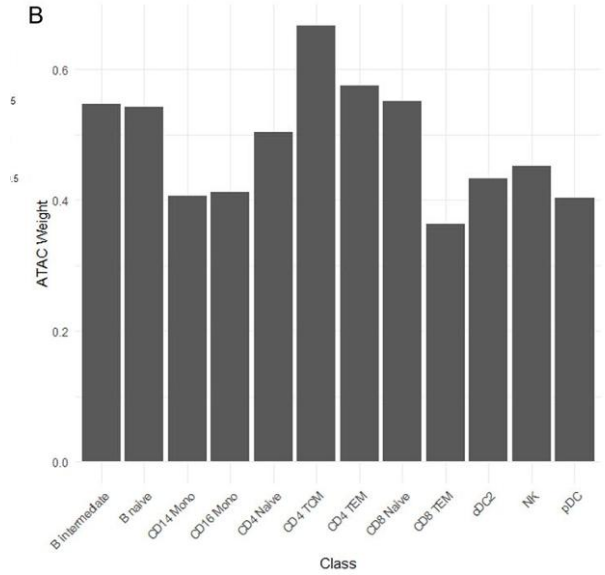
Pearson correlations of **scVI RNA + ATAC embeddings**
Multinomial LASSO coefficients for each cell type

Overall correlations were low, **indicating minimal shared structure**.
98.9% of cross-modality correlations fell between –0.3 and 0.3.

ATAC Contribution by Cell Type:
Highest ATAC association:CD4 TCM (66.6%),CD4 TEM (57.4%),CD8 naïve(55.1%)
Lowest ATAC association: CD8 TEM (36.3%), pDC (40.3%), cDC2 (43.3%)

- T cell subtypes (except CD8 TEM) show strong ATAC-driven signal.
- ATAC and RNA embeddings are largely independent, with ATAC providing unique subtype-specific information





ATAC does not confer an advantage in neuronal cell annotation

UMAP Visualization

Adding ATAC to scVI embeddings **creates more visually distinct clusters**, especially for inhibitory and excitatory neurons. Despite clearer separation, this did not translate into better classification.

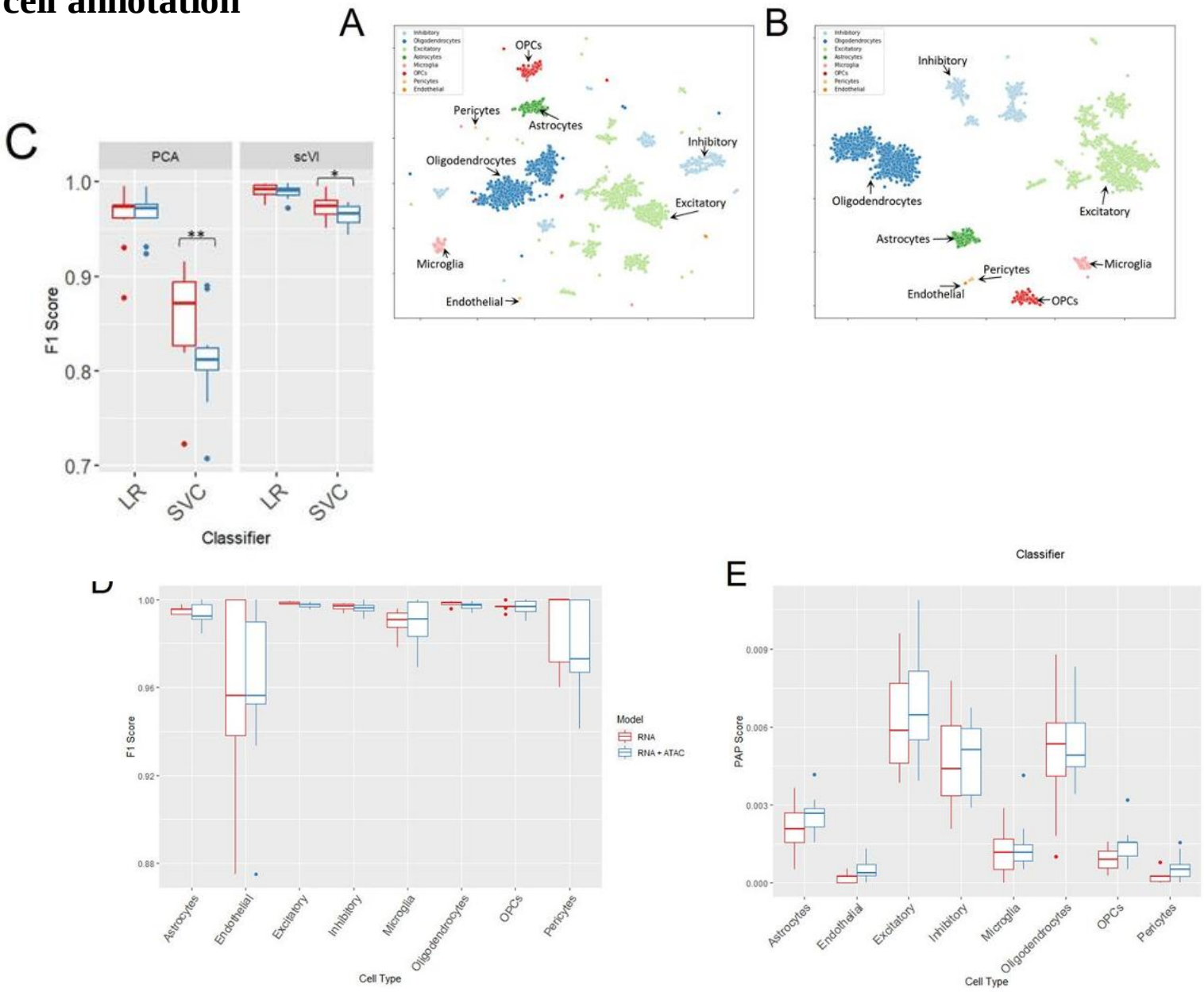
Classification Performance

Logistic Regression (LR**): No significant difference between** RNA-only and RNA+ATAC.SVM models: Performance **decreased when ATAC was added.** Largest drop: PCA-SVM

Cell-Type–Specific Effects

For the best model (scVI-LR), ATAC **produced no significant** changes in F1 or PAP across all neuronal subtypes.

- **Although ATAC improves visual separation of neuronal populations, it does not enhance annotation accuracy. For neuronal tissues, RNA alone performs as well or better.**



ATAC Utility Depends on Model, Cell Type, and Task

- ATAC + RNA improved annotation performance for PBMCs **only when using scVI**.
- **Largest gains** observed in **T cell subtypes**; **smallest gains** in dendritic cells.

Dimensionality Reduction Influences ATAC Utility

- **UMAP cluster separation improved** with ATAC **in both PBMCs and neuronal cells, but these visual improvements** did not match classification performance. PCA showed strong visual separation gains but no classification improvements.

- Explanation:
- UMAP preserves topology, not true geometric distances → distances not directly interpretable.
- Sensitive to parameters (e.g., n_components, min_dist), leading to arbitrary differences.

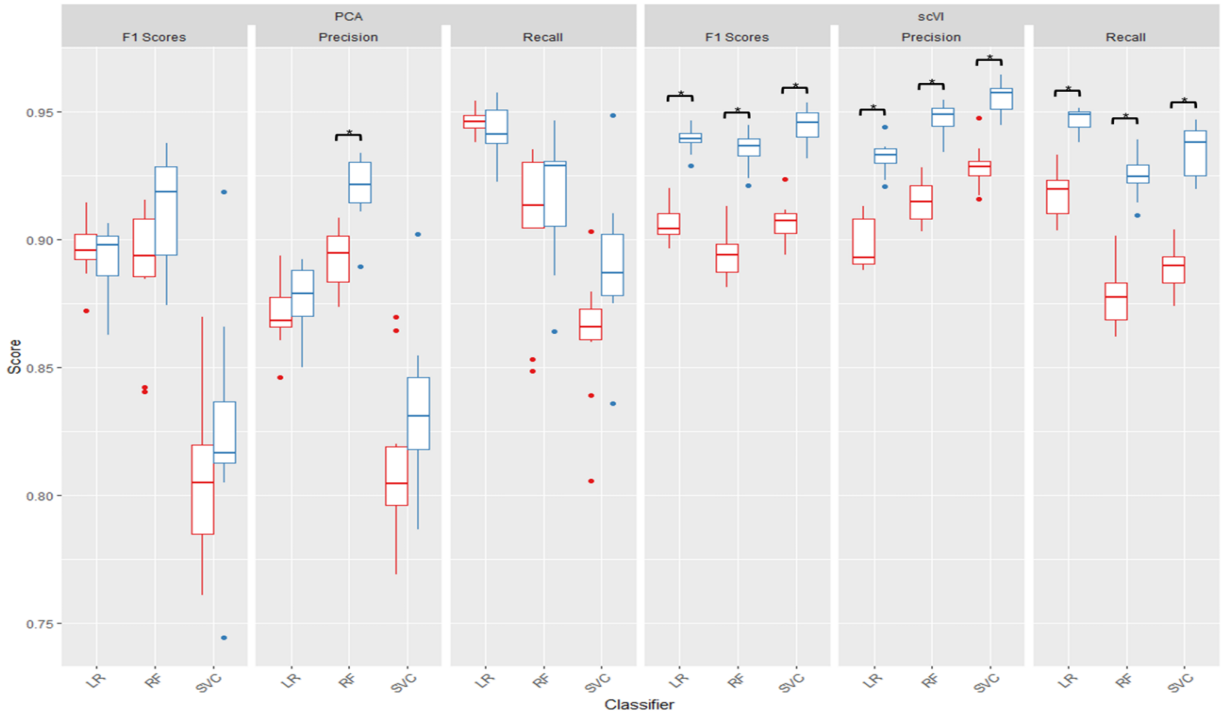
- Silhouette scores improved only in feature spaces that **preserve real distances** (PCA/scVI), not UMAP.

Why scVI Outperforms PCA

scVI (autoencoder) captures **non-linear structures** in ATAC, retaining more information.

PCA captures only **linear variance** and explained **<12% variance** per sample.

- Therefore, ATAC benefits appear **only when embeddings are expressive enough** (scVI).



Classifier Choice Does Not Change ATAC Utility

ATAC-driven improvements **were consistent across RF, LR, and SVM**.

Linear models performed similarly to non-linear models.

- Linear models may be preferred when interpretability and simplicity matter more than small accuracy gains.

Effect of ATAC on PBMC Annotation Quality

ATAC improvements were cell-type dependent

- Largest gains in **T cell subtypes**

Why T cells benefit most

- Presence of **poised/bivalent chromatin states**
- Epigenetic poising enables rapid gene activation before RNA changes
- Explains reduced misclassification (e.g., **CD4 TCM vs CD4 naïve**) when ATAC added
- **CD8 memory cells:** naïve-like RNA but effector-like ATAC → ATAC adds unique signal

Variability and limitations

- High variance in **CD4 TEM** improvements due to low sample frequency
- Some T-cell subtypes (e.g., **CD8 TEM**) show weaker ATAC associations
- Distal regulatory modifications may also contribute to signals captured by ATAC

Effect of ATAC on Neuronal Cell Type Annotation

Why do dendritic cells show minimal improvement

pDC: small added benefit due to already excellent RNA performance

cDC2: minimal ATAC benefit → ATAC signals less informative for dendritic subtypes

➢ Biological rationale: Dendritic cells **lack epigenetic poising**, unlike B/T cells.

No improvement with ATAC features

LR-scVI (best model): **no gains** for any neuronal cell type

SVM: ATAC **significantly worsened** performance (PCA + scVI embeddings)

Why ATAC may fail in neuronal data

The dataset includes **15 donors + two health states (healthy vs AD)**
→ high inter-individual + disease variation dilutes ATAC cell-type signals

SVM overfitting: ATAC likely adds **noise > signal**, amplified in non-linear models

Neuronal cells have **less dynamic epigenetic diversity** vs immune cells
→ fewer ATAC-detectable subtype differences

Cell Annotation Limitations

- **Manual cluster annotation issues**

  Depends on **non-standard marker gene lists**

  Does not incorporate **negative markers** or **multi-modal markers** (e.g., protein)

  Less reliable than **FACS-sorted gold-standard** annotations

- **Impact on model evaluation**

  Reported accuracy reflects limitations of **manual ground truth**, not true model performance

- **Dataset limitations**

  The study used **only two datasets** → limits generalizability for PBMCs and neuronal cells

  Larger, diverse datasets are needed to **validate and extend** these findings

Embedding Methods

- **Future work should explore improved embedding approaches** to better represent ATAC features, as model performance is highly dependent on the embedding method. (**deep adversarial learning)**
- Because autoencoders can overfit noise in sequencing data, **denoising models** may enhance robustness, especially given noise from **dropout, overdispersion, and zero inflation** in single-cell data.
- Additional strategies include developing **co-embedding methods to reduce redundancy** between modalities, such as **Multi-omic Factor Analysis (MOFA)**, which extends PCA to jointly model multiple omics layers.

Conclusion

This study showed that integrating **ATAC with RNA embeddings** generated using **scVI autoencoders** significantly improves **supervised annotation quality and prediction confidence** in **PBMC datasets** across **both linear (LR) and non-linear (RF, SVM) classifiers.**

Major performance gains were driven by enhanced discrimination of **fine-grained cell subtypes**, especially **T-cell subpopulations**.

No significant improvements were observed for **neuronal cell types, highlighting tissue-specific behavior.**